Module - 3

Road Safety in Transport Planning and Geometric Design

<u>Road User / Human Characteristics</u>

The human element plays a key role in road safety since pedestrians, cyclists, drivers, and passengers interact with traffic. Their characteristics can be grouped into four main categories:

a) *Physical Characteristics*

- Include vision, hearing, strength, and general physical fitness.
- Vision is the most important: visual clarity, field of vision (up to 10° for sharp focus, 30° for clear sight, and up to 200° for peripheral vision), glare resistance, and depth perception (the ability to judge how far away things are and how close) affect safety.
- Hearing also helps in detecting horns and sirens.
- Physical limitations such as age, fatigue, or disability reduce efficiency.

b) *Mental Characteristics* (decide how well you CAN react**)**

- Knowledge, skill, experience, and intelligence influence how road users respond to traffic.
- Reaction to traffic rules, regulations, and unexpected situations depends on mental alertness.
- Reaction time (PIEV theory – Perception, Intellection, Emotion, Volition) is crucial for safe driving.

c) *Psychological Characteristics* (decide how you CHOOSE to react***)***

- Attitudes, habits, and emotional stability affect road safety.
- Impatience, aggressiveness, or carelessness leads to risky driving.
- Distractions, stress, and lack of maturity reduce attentiveness to traffic conditions.

d) *Environmental Factors*

- Local conditions such as shopping centers, schools, and roadside activities affect user behavior.
- Weather conditions (rain, fog, night visibility) influence performance.
- Surrounding traffic stream and type of vehicles also modify human behavior.

Vehicle Characteristics

Vehicle features also strongly influence road safety and transport planning. These characteristics are classified into **static** (permanent) and **dynamic** (performance-related).

a) *Static Characteristics*

- **Vehicle Dimensions**: Width, height, and length affect lane width, parking, turning radius, and road capacity. IRC has set standards (e.g., max bus length 12 m, truck/trailer up to 18 m).
- **Weight of Vehicles**: Affects pavement design, gradients, and bridge loadings. IRC specifies maximum permissible gross weight and axle loads.
- **Turning Radius and Clearance**: Longer vehicles need larger turning radii; overhangs and wheelbase affect maneuverability.

b) *Dynamic Characteristics*

- **Speed**: Determines sight distance, curve design, lane width, capacity of roads, and intersection control. High speed reduces reaction time and increases crash severity.
- **Acceleration & Deceleration**: Depends on engine power, load, and road gradient. Influences overtaking ability and traffic flow.
- **Braking Characteristics**: Efficiency of braking systems (mechanical, air, hydraulic) determines stopping distance and safety. Poor brakes increase collision chances.
- **Stability & Comfort**: Wheelbase, suspension, and center of gravity affect skidding, overturning, and ride comfort.
- **Impact on Collisions**: Vehicle body design, bumpers, and crumple zones determine injury levels during crashes.

PIEV Theory of Reaction Time

The **PIEV Theory** explains the stages of a driver's total reaction time when faced with a stimulus (like seeing an obstacle) before taking an action (like braking).
It divides the reaction time into **four components**:

1. Perception Time (P)

- Time required for the driver's eyes/ears to sense an object or situation.
- Involves receiving signals through sense organs and transmitting them to the brain.
- Example: Noticing a pedestrian suddenly crossing the road.

2. Intellection Time (I)

- Time taken by the driver to understand and interpret the situation.

- Involves analyzing and comparing thoughts, deciding whether the object is dangerous.
- Example: Realizing that the pedestrian is directly in the vehicle's path.

3. Emotion Time (E)

- Time elapsed due to emotional and mental conditions like fear, anger, hesitation, or superstition.
- Varies from driver to driver and depends on the seriousness of the situation.
- Example: A nervous driver may take more time to react compared to a confident driver.

4. Volition Time (V)

- Time required for the driver to perform the final action, such as pressing the brake or turning the steering.
- It is the execution stage of the decision taken.
- Example: Applying the brakes to stop the vehicle.

Road Design (Geometric Design) and Safety Elements

Road design or geometric design means the way a road is planned and built with proper curves, slopes, width, intersections, drainage, and roadside safety. The aim is to ensure safety, smooth movement of vehicles, and comfort for all road users. If the design is not proper, it leads to confusion, reduced visibility, and accidents.

Important Road Design and Safety Elements

1. *Horizontal and Vertical Alignment*
    - Horizontal alignment refers to curves and bends on the road. Curves should be gentle so that drivers can see the road ahead. Sharp curves reduce visibility and increase chances of skidding or overturning.
    - Vertical alignment refers to slopes or gradients of the road. Gradual slopes are safe for movement, while very steep slopes may cause vehicles to roll back when going uphill or lose control when going downhill.

2. *Sight Distance*
    - Sight distance is the distance that a driver can clearly see on the road ahead. It is one of the most important safety factors.
    - Stopping Sight Distance (SSD): This is the minimum distance needed for a driver to stop safely after seeing an obstacle.
    - Overtaking Sight Distance (OSD): This is the distance required by a vehicle to overtake another vehicle safely without colliding with oncoming traffic.

- If the road does not provide enough sight distance, drivers may not react in time, leading to accidents.

3. *Lane Width and Shoulder Design*
   - Lane width: Proper lane width allows vehicles to move safely without side collisions. Wider lanes provide comfort and safety, while very narrow lanes increase chances of accidents.
   - Shoulders: These are extra spaces provided at the side of the road. They are used for emergency parking, for broken down vehicles, and also give space for vehicles that accidentally go out of the lane. Lack of proper shoulders increases accident risks.

4. *Intersection Design*
   - Intersections are places where two or more roads meet. These are complex points where chances of accidents are higher.
   - A properly designed intersection provides adequate turning radius, clear traffic signs, signals, and markings for safe movement.
   - These are circular intersections where vehicles move in one direction. Roundabouts are considered safer than normal intersections as they reduce head-on conflicts.
   - If intersections are not designed properly, they confuse drivers and increase the possibility of collisions.

5. *Cross Slopes and Drainage*
   - Cross slope: A small slope given across the road surface helps to drain rainwater quickly. This prevents water from collecting on the road.
   - Drainage: Proper drainage is important to keep the road safe and strong. If water collects, it causes skidding and a condition called hydroplaning where vehicles slide on water. It also damages the pavement over time.

6. *Roadside Design and Clear Zones*
   - Clear zones: These are open spaces provided at the edges of the road without fixed obstacles. They give a chance for vehicles that go out of control to recover safely.
   - Safety barriers: Guardrails, crash barriers, and crash cushions are placed at dangerous locations like sharp curves, bridges, or steep slopes to reduce the severity of accidents.
   - Without these facilities, roadside accidents can become more serious and fatal.

Road design and safety elements such as alignment, sight distance, lane width, shoulders, intersection design, cross slopes, drainage, and roadside safety features are essential in reducing accidents and ensuring comfort for drivers. A well-designed road provides safe and efficient travel, while poor design increases risks and endangers road users.

<u>Redesigning Junctions</u>

A junction is the point where two or more roads meet. It is one of the most critical parts of the road network because many conflicts happen here between vehicles, pedestrians, and cyclists. Junctions often face congestion and have high accident rates. Redesigning junctions is necessary to improve both safety and efficiency.

Safety Issues at Junctions

1. ***Conflicting Movements***
   - Vehicles coming from different directions may cross each other's path.
   - Chances of head-on, side, and rear-end collisions increase.

2. ***Insufficient Sight Distance***
   - Drivers may not clearly see approaching vehicles or pedestrians due to poor layout or obstructions.

3. ***High Speed of Vehicles***
   - At highways or wide urban roads, vehicles enter junctions at high speed, leading to severe crashes.

4. ***Poor Turning Facilities***
   - Small turning radius forces sharp turns, increasing the risk of overturning or side collisions.

5. ***Inadequate Pedestrian Facilities***
   - Pedestrians often cross between moving traffic if there are no zebra crossings, signals, or overpasses.

6. ***Poor Lighting and Signage***
   - Lack of lights or proper signboards causes confusion, especially at night or during bad weather.

Improvements by Redesigning Junctions

1. ***Channelization of Traffic***
   - Use of traffic islands, medians, and markings to guide vehicles into correct lanes.
   - Reduces conflicts between left-turning, right-turning, and straight-moving traffic.

2. ***Providing Proper Turning Radius***
   - Enlarging junction corners helps smooth turning movements.
   - Reduces vehicle skidding and improves comfort.

3. ***Roundabouts***
   - Circular intersections make vehicles slow down and move in one direction.
   - Reduces the chances of head-on and right-angle crashes.

4. ***Grade Separation***
   - Flyovers and underpasses separate traffic streams at different levels.
   - Removes direct conflict points and improves capacity of busy junctions.

5. ***Traffic Signals and Signage***
   - Proper signal timing gives each direction safe time to move.
   - Stop lines, road markings, and clear signboards help in regulating flow.

6. ***Pedestrian and Cyclist Facilities***
   - Zebra crossings, pedestrian signals, subways, overbridges, and cycle lanes ensure safe movement of non-motorized users.

7. ***Improved Visibility and Lighting***
   - Removal of obstructions near junctions increases sight distance.
   - Street lighting at junctions improves night-time safety.

Cross Section Improvements

The cross section (Fig 1) of a road means the arrangement of different elements across its width, such as carriageway, shoulders, medians, footpaths, and side drains. A properly designed cross section ensures safety, comfort, drainage, and smooth traffic flow. Improvements in cross section are often required when traffic increases, accidents occur frequently, or when the road condition is not suitable for present needs.

Key Cross Sectional Elements of Roads

1. ***Carriageway***
   - The main portion of the road surface used by moving vehicles.
   - Its width depends on traffic volume and type of road.

2. ***Shoulders***
   - Provided on both sides of the carriageway.

- Used for emergencies, parking of broken-down vehicles, and pedestrian use in rural roads.

3. ***Medians or Dividers***
   - Separating strip between two directions of traffic.
   - Improves safety by reducing head-on collisions.

4. ***Footpaths and Cycle Tracks***
   - Provided in urban areas for pedestrians and cyclists.
   - Prevents mixing of slow and fast traffic.

5. ***Side Drains***
   - Arranged along road edges to carry away rainwater.
   - Essential for maintaining road life and preventing waterlogging.

6. ***Roadside Slopes and Safety Features***
   - Slopes and embankments form part of the road cross section.
   - Guardrails and barriers are added to protect vehicles from accidents at the edges.

Safety Issues in Cross Sections

1. ***Narrow Carriageway***
   - Causes congestion and side collisions, especially with mixed traffic.

2. ***Lack of Shoulders***
   - Vehicles in emergencies have no safe space to stop, leading to accidents.

3. ***Absence of Median or Divider***
   - Opposite traffic flows are not separated, leading to head-on collisions.

4. ***Poor Drainage***
   - Water collects on the surface, causing skidding and pavement damage.

5. ***No Pedestrian or Cycle Facilities***
   - Pedestrians and cyclists mix with fast traffic, increasing accident risks.

6. ***Unsafe Roadside Slopes***
   - Steep or unprotected side slopes can cause rollovers if vehicles leave the carriageway.

Methods of Cross Section Improvements

1. ***Widening of Carriageway***
    - Provides more space for vehicles, reduces congestion, and allows safe overtaking.
    - Standard lane width of 3.5 m ensures stability on high-speed roads.

2. ***Improved Shoulder Design***
    - Paved or stabilized shoulders provide safe space for emergency parking and recovery.
    - Protects pavement edges and adds stability for vehicles at the road edge.

3. ***Provision of Medians/Dividers***
    - Separates opposite traffic flows, preventing head-on collisions.
    - Median barriers (concrete, cable, or guardrails) reduce crash severity and control vehicle movement.
    - On busy roads, medians help restrict unsafe mid-block turns and allow safer U-turns.

4. ***Better Drainage Facilities***
    - Side drains and proper cross slopes quickly remove water.
    - Prevents skidding, hydroplaning, and increases pavement life.

5. ***Pedestrian and Cycle Tracks***
    - Separate footpaths and cycle lanes protect vulnerable road users.
    - Reduces accidents involving slow-moving and non-motorized traffic.

6. ***Roadside Safety Improvements***
    - Flattening of slopes, and installation of guardrails, crash barriers, or cable barriers.
    - Reduces severity of run-off-road crashes.

7. ***Improved Pavement Surface***
    - Smooth, skid-resistant pavement improves control and riding comfort.
    - Reduces wear and tear of vehicles and increases overall safety.

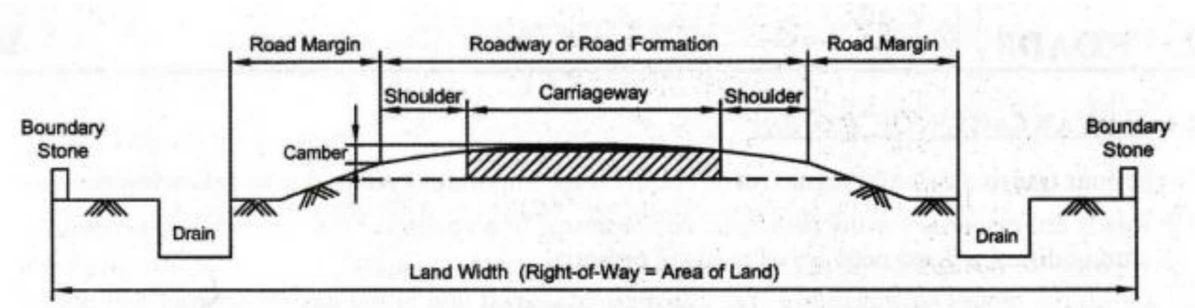
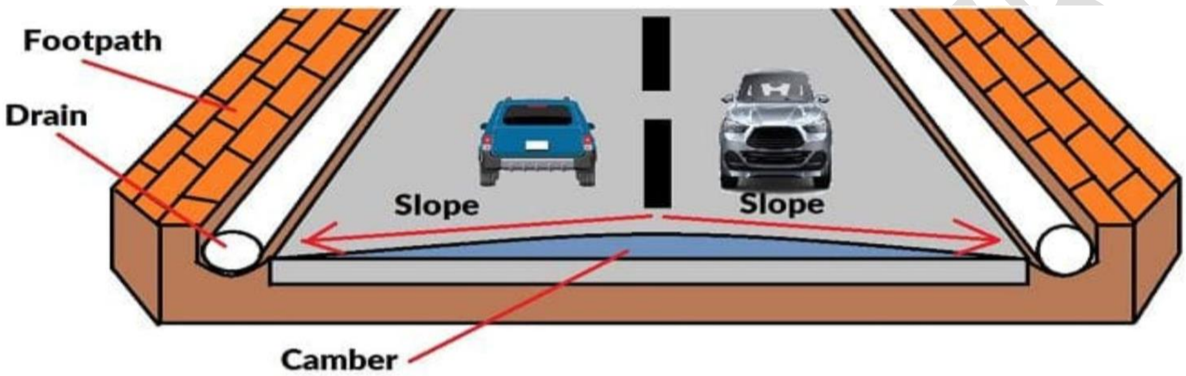
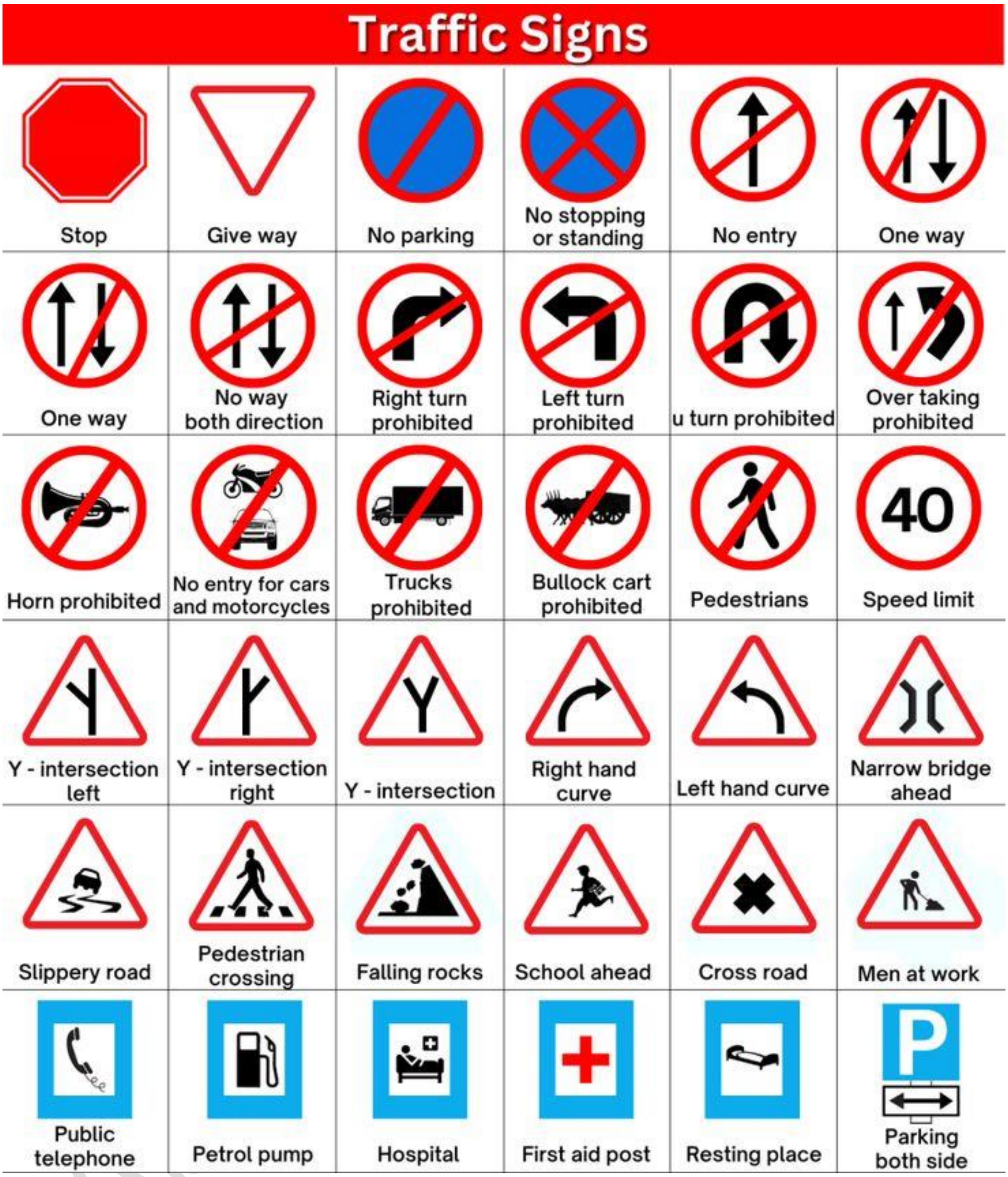


FIG. 1 CROSS-SECTION DETAILS OF A ROAD




Traffic Signs
Stop
Give way
No parking
No stopping or standing
No entry
One way
One way
No way both direction
Right turn prohibited
Left turn prohibited
u turn prohibited
Over taking prohibited
Horn prohibited
No entry for cars and motorcycles
Trucks prohibited
Bullock cart prohibited
Pedestrians
40
Speed limit
Y - intersection left
Y - intersection right
Y - intersection
Right hand curve
Left hand curve
Narrow bridge ahead
Slippery road
Pedestrian crossing
Falling rocks
School ahead
Cross road
Men at work
Public telephone
Petrol pump
Hospital
First aid post
Resting place
P
Parking both side

Traffic Control Devices

Traffic control devices are tools used to regulate, guide, and warn road users. They include **signs, signals, pavement/road markings and islands** which communicate important information for safe and efficient road use. Properly designed and maintained devices reduce conflicts and accidents, while missing or unclear devices may lead to confusion, violations, and crashes.

1. Traffic Signs

Traffic signs provide **regulatory, warning, and guidance information**.

(a) Regulatory Signs

- Tell drivers the laws and rules they must obey.
- Examples: STOP, YIELD, NO ENTRY, NO U-TURN, SPEED LIMIT signs.
- Importance: A missing or obscured STOP sign at an intersection can lead to dangerous right-angle (T-bone) collisions.

(b) Warning Signs

- Give advance notice of potential hazards on the road.
- Examples: Curve ahead, pedestrian crossing, school zone, slippery road.
- Importance: Without warning, drivers may approach curves too fast or fail to slow down for pedestrians, causing accidents.

(c) Guide Signs

- Provide navigation help such as directions, distances, and route numbers.
- Examples: Highway route markers, distance boards, city direction signs.
- Importance: Absent or misleading guide signs can cause sudden lane changes or turns, increasing crash risk.

2. Traffic Signals

Traffic signals control the **movement of vehicles and pedestrians** at intersections, especially in busy areas.

Functions of Traffic Signals

- ***Assign Right-of-Way*** – Signals reduce conflicts by giving each movement a fixed time to proceed.
- ***Improve Pedestrian Safety*** – Provide separate pedestrian crossing phases.
- ***Regulate Flow*** – Maintain order in high-volume or complex intersections.

- ***Adapt to Conditions*** – Signal timing (green, yellow, red duration) must match traffic flow to avoid unsafe behaviors.

Safety Issues in Traffic Signals

- If a signal light is unclear, missing, or has poor timing, drivers may get confused.
- Yellow Light Dilemma Zone – If too short, drivers may brake suddenly or run red lights, causing rear-end or side collisions.
- Example: At a junction with short yellow duration, fast vehicles may be caught between stopping or crossing, leading to crashes.

3. Pavement/road Markings

Pavement markings are lines, arrows, symbols, or words painted on the road surface. They guide drivers, regulate lane usage, and improve discipline.

Functions of Pavement Markings

a) Centerlines and Lane Dividers

- Separate lanes of traffic and help maintain proper lane discipline.
- Example: Double solid lines prohibit overtaking.

b) Crosswalks and Stop Lines

- Indicate where vehicles must stop at signals or signs.
- Provide safe crossing paths for pedestrians.

c) Arrows and Wording on Pavement

- Direct traffic movement such as left turn only, straight only.
- Prevent confusion at intersections.

d) Edge Lines

- Mark the edges of the carriageway, especially useful at night or in poor visibility.
- Help prevent vehicles from drifting off the road.

Safety Issues in Pavement Markings

- Faded markings can confuse drivers and cause lane drifting or side-swipe collisions.
- Unmarked pedestrian crossings near schools or bus stops put pedestrians at risk, as drivers may not yield.

e) Traffic Islands

Traffic islands are raised areas constructed within a roadway to control vehicle movement, separate traffic streams, and provide safe zones for pedestrians. They improve order, reduce conflict points, and increase safety at intersections and busy roads.

Types of Traffic Islands (Based on Function)

a) Divisional Islands

- Separate opposing traffic flows on highways and multi-lane roads.
- Reduce head-on collisions by dividing the road into two one-way carriageways.
- Must be wide enough and have kerbs high enough to prevent vehicles from entering.

b) Channelizing Islands

- Guide vehicles into proper lanes at intersections.
- Reduce conflict areas between traffic streams.
- Control angles of merging and diverging traffic.
- Provide space for installing signs and signals.
- Can also serve as refuge islands for pedestrians.

c) Pedestrian Loading and Refuge Islands

- Provided near bus stops and crosswalks to protect passengers and pedestrians.
- Offer a safe waiting area before crossing multi-lane roads.

d) Rotary (Central) Islands

- Large central islands used in roundabouts.
- Convert crossing traffic into weaving movement with reduced conflict.
- Improve efficiency of traffic handling at intersections with balanced flow.

Safety Issues in Islands

- Poorly designed or placed islands may confuse drivers.
- Inadequate size may fail to reduce conflict points.
- Lack of visibility or proper markings may cause collisions.

Functions/Improvements with Traffic Islands

- Reduce conflict areas at intersections.
- Provide safe pedestrian refuge zones.
- Channelize vehicles into correct lanes.
- Improve traffic capacity and reduce delays.

- Enhance safety by preventing head-on collisions and uncontrolled turns.

Traffic Calming Measures

Traffic calming refers to the use of physical design features and management strategies on roads to reduce vehicle speeds, control driver behavior, and improve safety for pedestrians and cyclists. It is widely applied in residential areas, near schools, and high-pedestrian zones. The aim is not only to reduce accidents but also to create a safer and more livable road environment.

Key Traffic Calming Measures

1. *Speed Humps/Tables and Speed Breakers*
    - Raised portions across the road designed to slow vehicles.
    - Speed humps/tables: Longer, smoother, often combined with pedestrian crossings (safe near schools and housing areas).
    - Speed breakers: Shorter, steeper, commonly used in India but must follow design standards (IRC guidelines) to avoid accidents.
2. *Rumble Strips*
    - Series of raised strips across the roadway that create vibration and noise.
    - Warn drivers to reduce speed before hazardous areas like sharp curves, junctions, or toll plazas.
3. *Road Narrowing (Chicanes and Lane Width Reduction)*
    - Intentional narrowing of the carriageway or creating curves (chicanes) to force vehicles to slow down.
    - Narrower lanes psychologically reduce speeding.
4. *Raised Pedestrian Crossings*
    - Crosswalks elevated to the level of the footpath.
    - Makes pedestrians more visible and forces vehicles to slow while crossing.
5. *Traffic Islands and Refuge Islands*
    - Provide safe waiting space for pedestrians in the middle of wide roads.
    - Also channelize vehicle flow and reduce conflict points.
6. *Roundabouts*
    - Replace signalized intersections with circular flow.
    - Reduce speed of entry and minimize head-on and right-angle collisions.

7. *Visual Measures (Signage and Road Surface Treatments)*
   - Use of coloured pavement, warning signs, road markings, and landscaping.
   - Provides psychological cues for drivers to slow down.

<u>Road Safety Furniture</u>

Road safety furniture are fixed devices installed on and along roads to improve safety, guide drivers, and minimize accident severity. They include signs, markings, barriers, reflectors, and pedestrian facilities. They are not structural parts of the road but play a major role in traffic regulation and accident prevention.

Important Road Safety Furniture

1. *Road Signs*
   - Essential: Provide drivers with instructions, warnings, and guidance about road conditions, laws, and hazards.
   - Influence on Safety: Prevents confusion, ensures lane discipline, and reduces crash risks by giving drivers advance information.

2. *Traffic Signals*
   - Essential: Control traffic flow at busy intersections and provide safe phases for vehicles and pedestrians.
   - Influence on Safety: Reduces collisions at junctions, improves pedestrian safety, and regulates movements during peak traffic.

3. *Pavement Markings*
   - Essential: Painted lines, arrows, stop lines, and crosswalks guide drivers on lane usage and stopping positions.
   - Influence on Safety: Improves lane discipline, prevents side collisions, and ensures safe pedestrian crossings.

4. *Crash Barriers / Guardrails*
   - Essential: Installed along highways, bridges, and embankments to prevent vehicles from leaving the carriageway.
   - Influence on Safety: Absorb impact, reduce crash severity, and prevent vehicles from falling into dangerous areas.

5. *Medians and Dividers*
   - Essential: Physically separate traffic moving in opposite directions.

- Influence on Safety: Prevent head-on collisions and provide safe waiting areas for pedestrians.

6. *Pedestrian Railings and Refuge Islands*
   - Essential: Protect pedestrians from directly crossing fast-moving traffic and provide safe waiting zones.
   - Influence on Safety: Reduces pedestrian accidents, especially on multi-lane roads and near schools or bus stops.

7. *Street Lighting*
   - Essential: Provides adequate illumination on roads, intersections, and crossings during night.
   - Influence on Safety: Improves night visibility, reduces accidents due to poor sight, and increases comfort for drivers.

8. *Reflectors, Cat's Eyes, and Road Studs*
   - Essential: Installed on pavements and road edges for guidance during night or fog.
   - Influence on Safety: Help drivers maintain lane position, especially on curves and low-visibility conditions.

9. *Rumble Strips / Speed Calming Devices*
   - Essential: Raised strips on roads to alert drivers and force reduction of speed.
   - Influence on Safety: Warn drivers before intersections, schools, and hazardous locations, reducing overspeed crashes.

10. *Overhead Signboards / Directional Boards*
    - Essential: Large boards at highways and junctions for advance information about destinations, routes, and exits.
    - Influence on Safety: Prevents sudden lane changes, reduces driver stress, and ensures smoother flow of vehicles.